INDIA INNOVATES 2026

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Tech Titans

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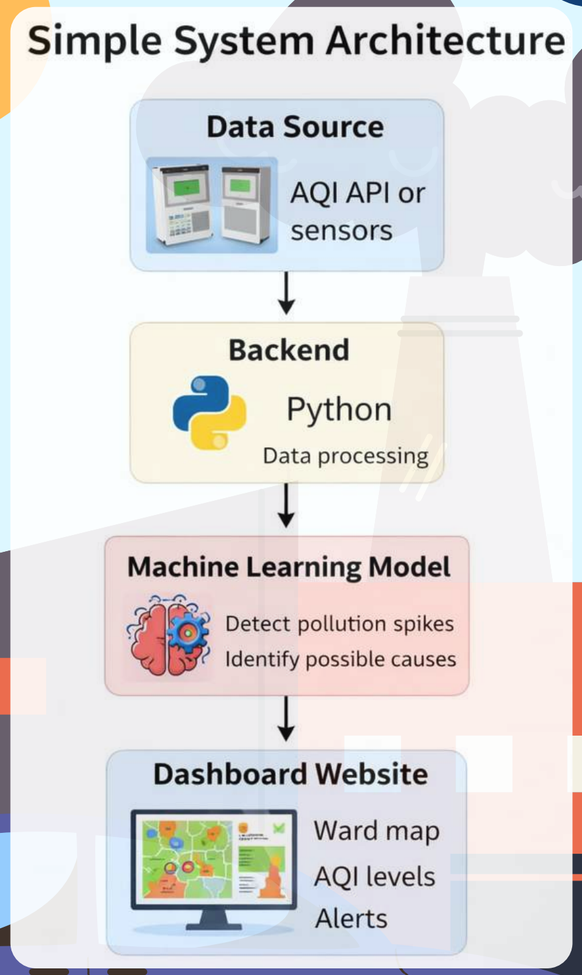
Problem Statement

Current urban air quality monitoring relies on **broad city-wide averages** that fail to capture **"hyper-local" pollution spikes**. This lack of **granular data** prevents city administrators from identifying specific sources—like construction sites or biomass burning—and leaves citizens unaware of the **immediate health risks** in their specific neighborhood or ward. We are fighting a dynamic, street-level enemy with **blind, city-wide averages**. Our cities need more than **passive monitoring**; they require a **hyper-local intelligence system.**

SOLUTION

Our solution is an **end-to-end,** ward-wise **AI dashboard** that bridges the gap between raw environmental data and **immediate action.**

1. ***Detect***: Deploys a **dense network of low-cost IoT sensors** to capture PM2.5 and PM10 metrics at the **street level.**
2. ***Analyze***: Uses a **Machine Learning classification model** to calculate particle ratios, instantly **identifying localized sources** (e.g., Construction Dust vs. Biomass Burning).
3. ***Act***: Automatically generates **targeted policy recommendations** for city administrators and **real-time health advisories** for local citizens based on the detected source



Architecture

1. Capture: Hardware sensors (SPS30) actively sample *PM2.5* and *PM10* particles at the street level.
2. Transmit: An *ESP32* microcontroller packages this data into *JSON* and pushes it to a Firebase Cloud Database via *MQTT*.
3. Process: A *Python* backend fetches the live data and feeds it into a *Scikit-Learn Machine Learning model*.
4. Act: The ML model calculates the particle size ratio to identify the pollution source, triggering our logic engine to instantly update the Web Dashboard.

Ward	PM2.5	PM10	Ratio	Detected Source	Alert Sent
Ward 12	87 µg/m³	210 µg/m³	0.41	Construction Dust	✅ Admin SMS
Ward 07	134 µg/m³	160 µg/m³	0.84	Biomass Burning	✅ Citizen SMS
Ward 19	95 µg/m³	175 µg/m³	0.54	Heavy Traffic	✅ Reroute advisory

Outcome

Shifts the city from passively monitoring air quality to actively catching and stopping emission sources in real-time.

FEATURE / USP

Standard dashboards tell you the air is toxic; our system tells you why and what to do about it.

Context-Aware ML: We don't just measure dust; we analyze **PM2.5/PM10 ratios** and weather contexts to **pinpoint the exact local offender** (e.g., stopping an illegal garbage fire).

Action-Driven Output: Removes human delay. When a hotspot forms, the system doesn't wait—it instantly generates **localized health SMS alerts** for citizens and **specific mitigation protocols** (like halting local construction) for administrators.

Cost vs. Legacy Systems: *Our per-node cost is **~95% cheaper** than a government CAAQMS station, enabling dense **ward-level coverage** that was previously impossible at city scale.*

Coverage and Scaling

Air Quality Network Proposal

Coverage — What 1 Node Can Do

1 node monitors air quality in a ~300–500m radius

1 ward needs 4–6 nodes for complete coverage

Cost per ward: ₹39,000

Component	Purpose	Est. Cost
SPS30 Sensor	Measures PM2.5 & PM10	₹4,500
ESP32 Microcontroller	Data processing & WiFi	₹400
Weatherproof Enclosure	Outdoor protection	₹800
Power Supply Unit	Continuous operation	₹600
Wiring & Connectors	Assembly	₹200
Total per Node		~₹6,500

*Figures are initial estimates.

City-Scale Comparison

	Bengaluru	Delhi
Total Wards	225 (BBMP 2023)	250 (MCD)
Total Area	~741 km²	~603 km²
Avg. Ward Size	~3.3 km²	~5.6 km²
Nodes Needed per Ward	4-6	6-8
Total Nodes (Full City)	~1,125	~1,750
Total Deployment Cost	~₹73 Lakh	~₹1.1 Crore
Govt. CAAQMS equivalent	₹11 Crore	₹60+ Crore

TECHNOLOGY USED

Technology / Tool	Category	Contribution to System
ESP32 & SPS30	Hardware (IoT)	**Efficiency**: Provides highly accurate, low-cost particle counting with a self-cleaning mechanism for long-term field deployment.
Firebase / MQTT	Database & Comms	**Scalability:** Easily handles massive, rapid streams of real-time JSON data from thousands of ward sensors simultaneously.
Python & Scikit-Learn	Machine Learning	**Functionality:** Uses a Random Forest Classifier to process PM ratios, allowing the system to intelligently categorize pollution
React / Streamlit	Frontend Dashboard	**Functionality:** Delivers a live, interactive map interface ensuring administrators can visualize alerts and execute policies immediately.

Data-Driven Policy Enforcement

DON'T JUST MANAGE THE CITY'S POLLUTION

Eliminate it—ward by ward, source by source.

- Immediately deploy **water cannons on-site** (**Treated sewage water to conserve resources and maximize water efficiency**) and install **suppression nets**.
- Re-routing heavy vehicles from the ward, enforce temporary **one-way traffic**, and promote **last-mile EV delivery** solutions.
- **Precision Planting :** Deploying **pollution-absorbing trees** and **moss walls** in data-identified hotspots.
- If the **AQI value** lies in the **unhealthy range(>200)**, immediate **Inspection** must be conducted.
- **Providing Rank** for each ward weekly, according to AQI and **make people motivated** to control it.

FROM REAL-TIME DATA TO REAL-WORLD DEFENSE

References / Links

- SPS30: https://sensirion.com/products/catalog/SPS30
- CAAQMS Accuracy Study (CSIR-NPL) – https://www.nplindia.org
- CPCB Air Quality Monitoring – https://cpcb.nic.in
- PM2.5/PM10 Basics (EPA) – https://www.epa.gov/pm-pollution/particulate-matter-pm-basics
- BBMP Ward Data – https://bbmp.gov.in
- Delhi MCD Ward Data – [suspicious link removed]
- Scikit-Learn – https://scikit-learn.org
- Streamlit – https://streamlit.io
- Firebase – https://firebase.google.com/docs/database

THANK YOU